be defined much more comprehensively on the basis of energy bands as under:

A semiconductor is substance which has almost filled valence band and nearly empty conduction band with a very small energy gap (=1 eV) separating the two. Fig 3.18 shows the energy band diagrams of germanium and silicon respectively. It may be seen that forbidden energy gap is very small, being 1.1 eV for silicon and 0.7 eV for germanium. Therefore, relatively small energy is needed by their valence electrons to cross over to the conduction band and thus become free electrons. However, at this temperature, number of free electrons available is very small. Therefore, at room temperature, a piece of germanium or silicon is neither a good conductor nor an insulator. For this reason, such substances are called semiconductors.

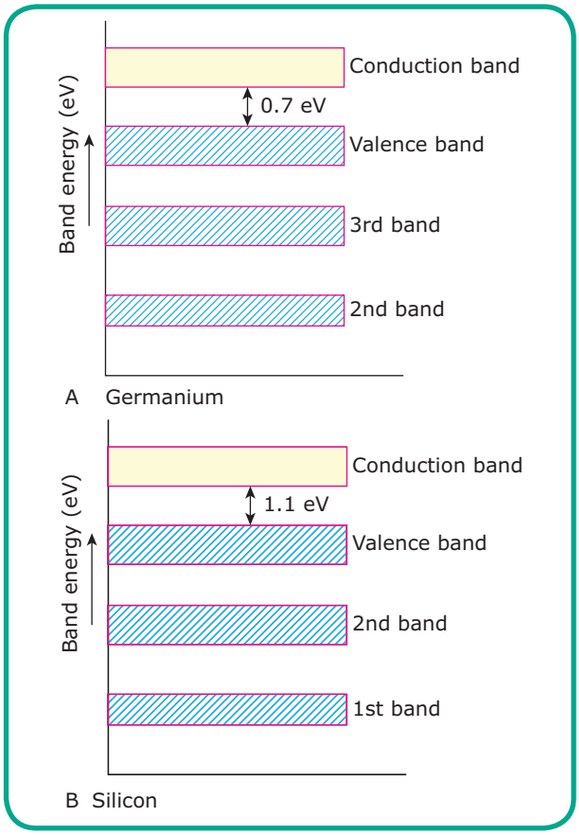


Figure3.18 Energy band description

The energy band description is extremely helpful in understanding the current flow through a semiconductor. Therefore, we shall frequently use this concept in our further discussion.

3.6.6 Hole Current

At room temperature some of the covalent bond in pure semiconductors breaks, setting up free electrons. Under the influence of electric field, these free electrons constitute electric current. At the same time, another current, the hole current also flows in the semiconductors. When the covalent bond is broken due to thermal energy, the removal of one electron leaves a vacancy, i.e., a missing element in covalent bond. This missing electron is called a hole, which acts as a positive charge. For one electron set free, one hole is created. Therefore, thermal energy creates hole-electron pairs. Hence, as many holes as free electrons are generated. The current conduction by holes can be explained as follows.

The hole shows a missing electron. Suppose the valence electron at L (Figure 3.19) has free electrons due to thermal energy.

This creates a hole in the covalent bond L. Now the hole becomes strong centre of attraction for the electron. So a valence electron (say at M) from nearby covalent bond comes to fill in the hole at L. This results in creation of hole at M. Another valence electron (say at N) in turn may leave its bond to fill the hole at M, thus creating a hole at N. Thus, the hole having positive charge has moved from L to N i.e., towards the negative terminal of supply. This constitutes the hole current.

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